

Power System

Power System

The diagram illustrates a power system with the following components and connections:

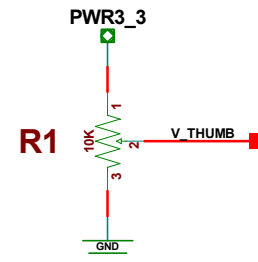
- Inputs:** RAW BATTERY (J0), PWR5E+ (J5), PWR3_3G (J11), PWR3_3 (J12).
- Outputs:** PWR5A+ (J10), PWR5E+ (J5), PWR3_3G (J11), PWR3_3 (J12), V_BAT (J22), V_3.3 (J18).
- Key Components:**
 - U1:** LT1935ES5 (Voltage Regulator)
 - D1, D2:** PMEG3020BEP (Diodes)
 - L1:** 10UH, 74489440100 (Inductor)
 - C1-C12, C22:** Various capacitors (100UF, 35V, X5R; 470PF, 50V; 22UF, 25V)
 - R1-R9, R17, R18:** Various resistors (10.0K, 20.0K, 12.4K, 33, 10K)
 - Q2, Q3, Q4:** ZXMN2F30FH, ZXMP3A13F (MOSFETs)
 - SW1:** Switch
 - Thumb Wheel:** V_THUMB (J18)

Thumb Wheel

The Thumb Wheel component is connected to the PWR3_3G input and the V_THUMB output.

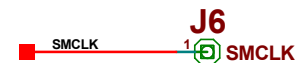
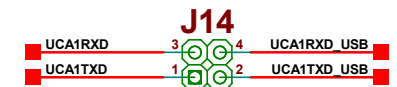
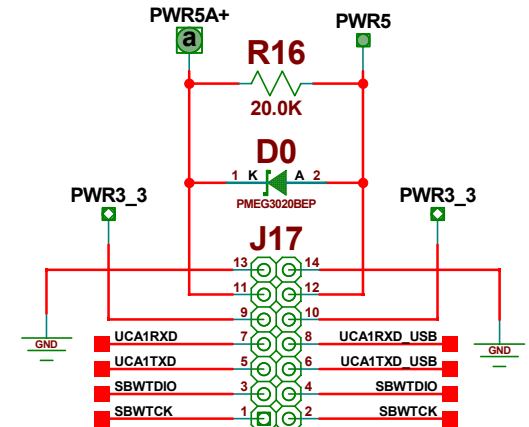
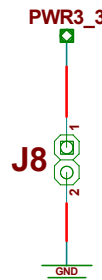
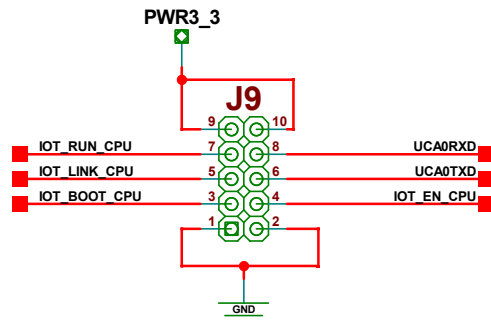
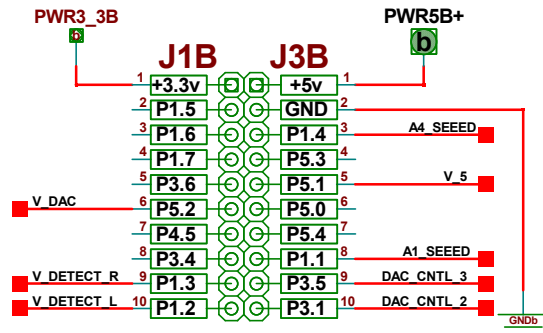
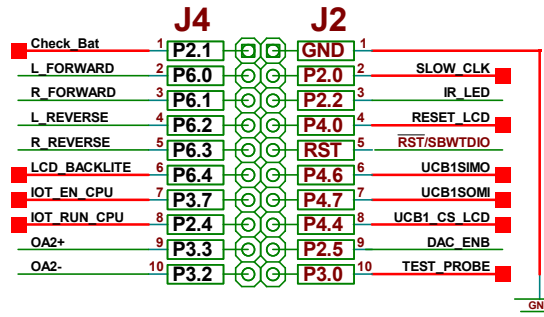
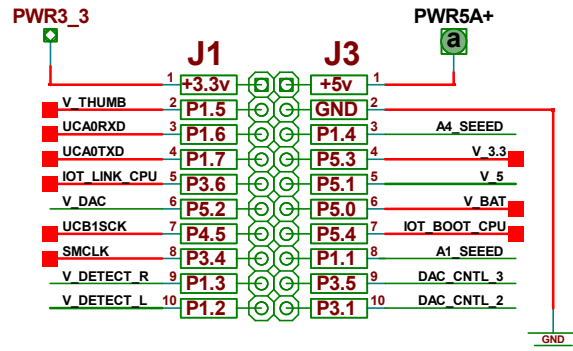
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NCSU					
DR:	REL:				
Jim Carlson					
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SUB SHEET TITLE:	Power Input				
SCALE:	SIZE:	DRAWING NUMBER:	SUB-SHEET:	SHEET:	REV:
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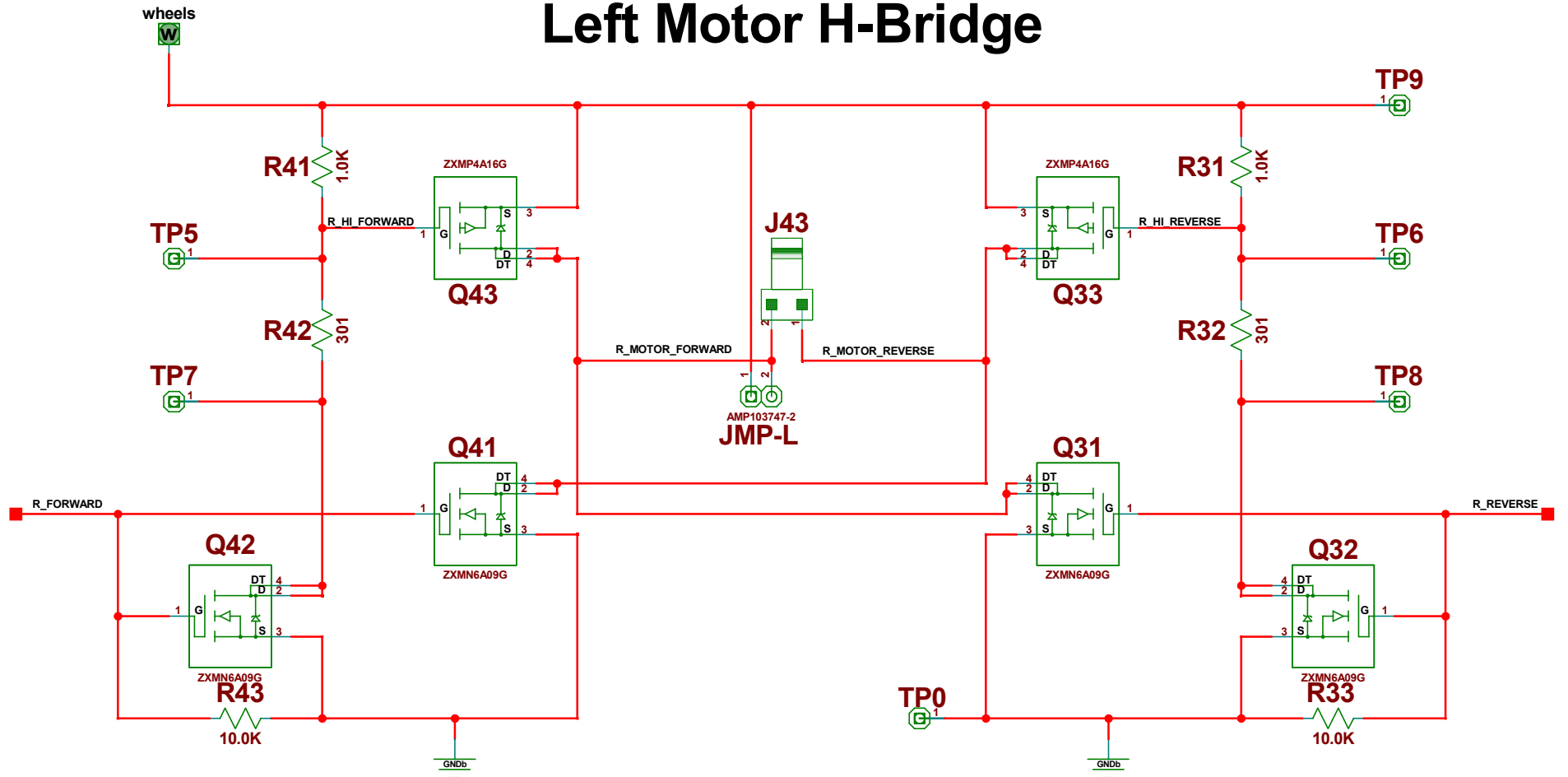
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SUB SHEET TITLE: Power Input					
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					01
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Interconnect



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DR: Jim Carlson	REL:	
FILENAME: FILE_NAME		
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SUB SHEET TITLE: Interconnect		
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SUB-SHEET: 1 OF 1	SHEET: 2 OF 8	REV: 01
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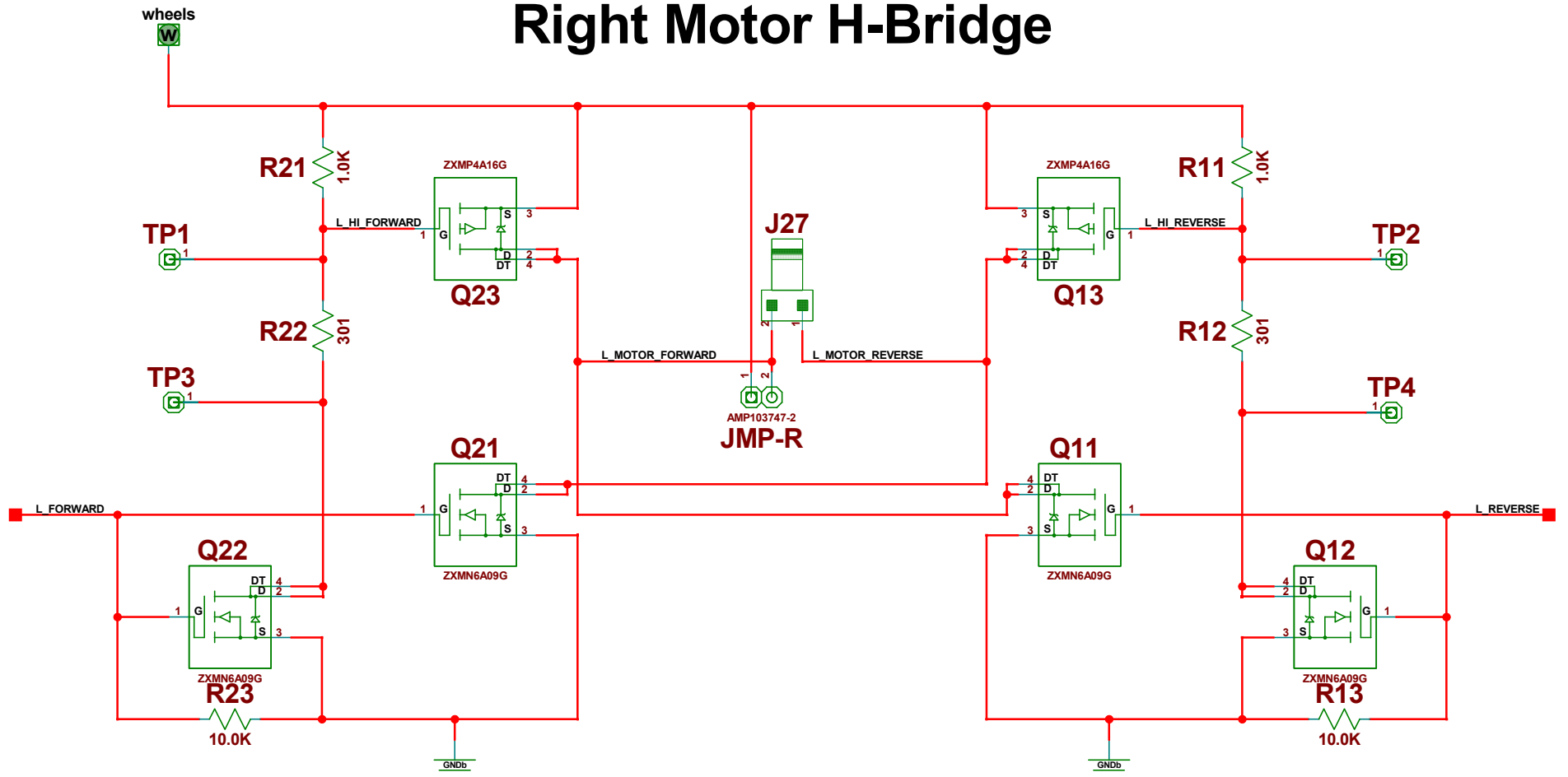
Left Motor H-Bridge



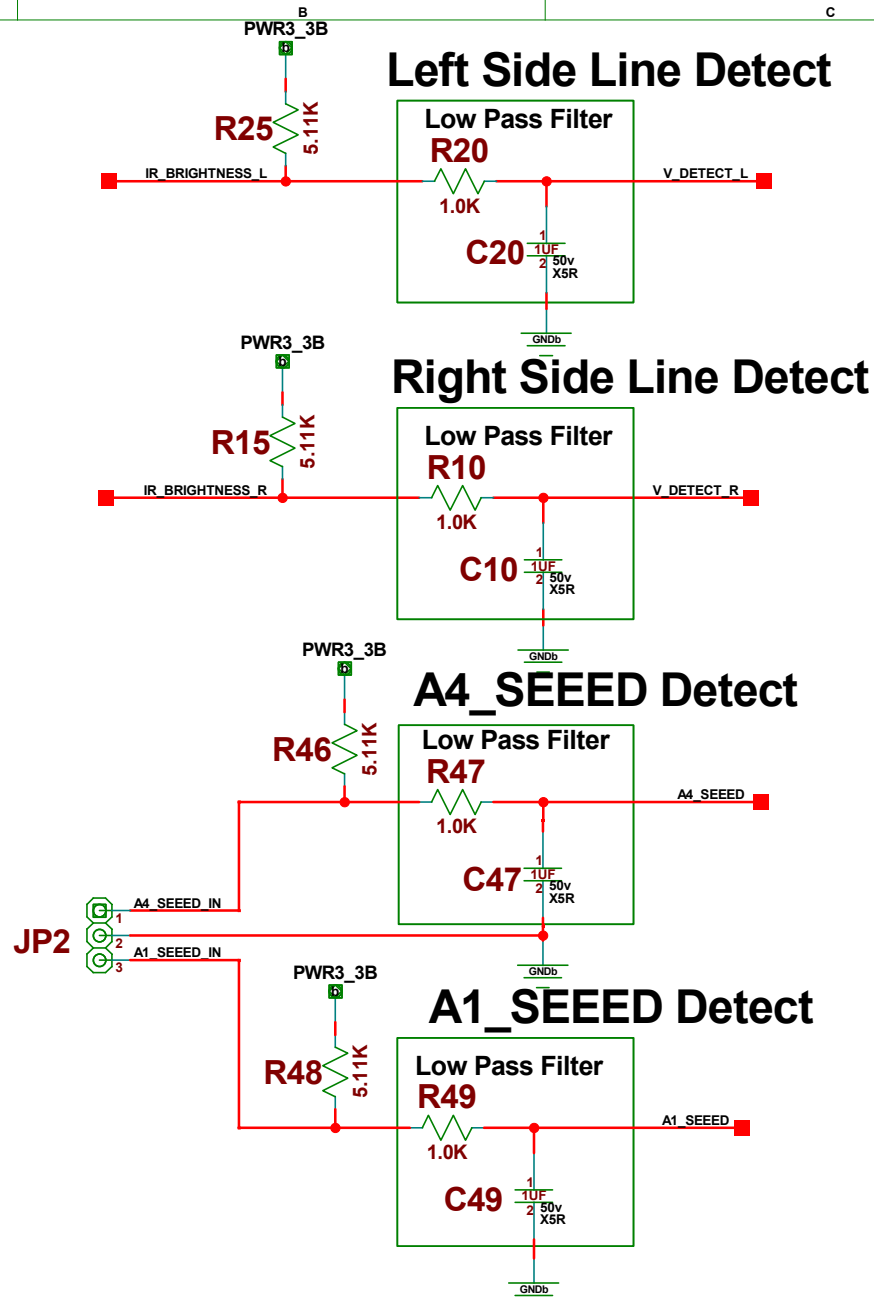
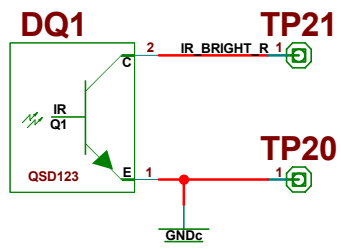
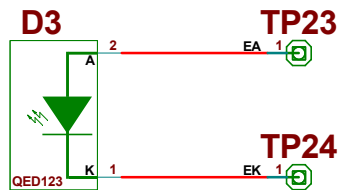
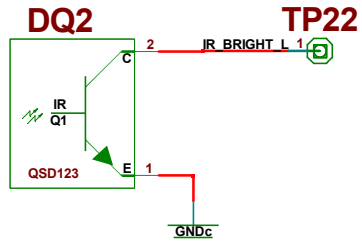
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DR: Jim Carlson		REL:			
FILENAME:		FILE_NAME			
TITLE: ECE306 Fall 2025					
SUB SHEET TITLE: Left Motor H-Bridge					
SCALE: 1:1	SIZE: B	DRAWING NUMBER: ECE306	SUB-SHEET: 1 OF 1	SHEET: 3 OF 8	REV: 01

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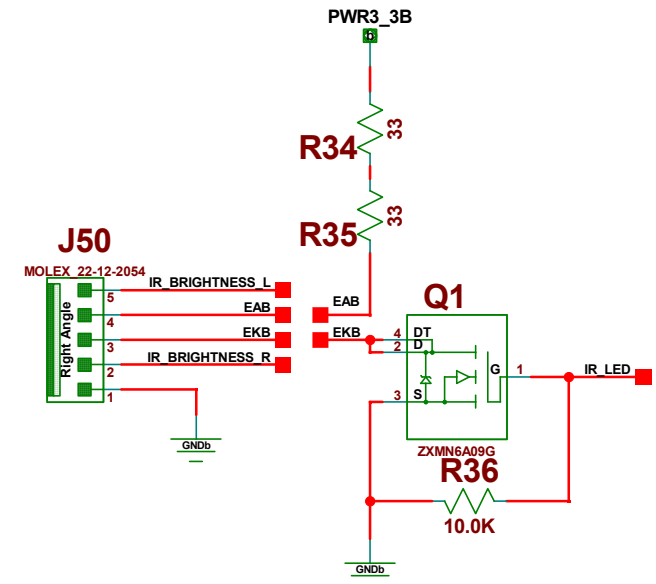
Right Motor H-Bridge



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DR: Jim Carlson		REL:			
FILENAME:		FILE_NAME			
TITLE: ECE306 Spring 2025					
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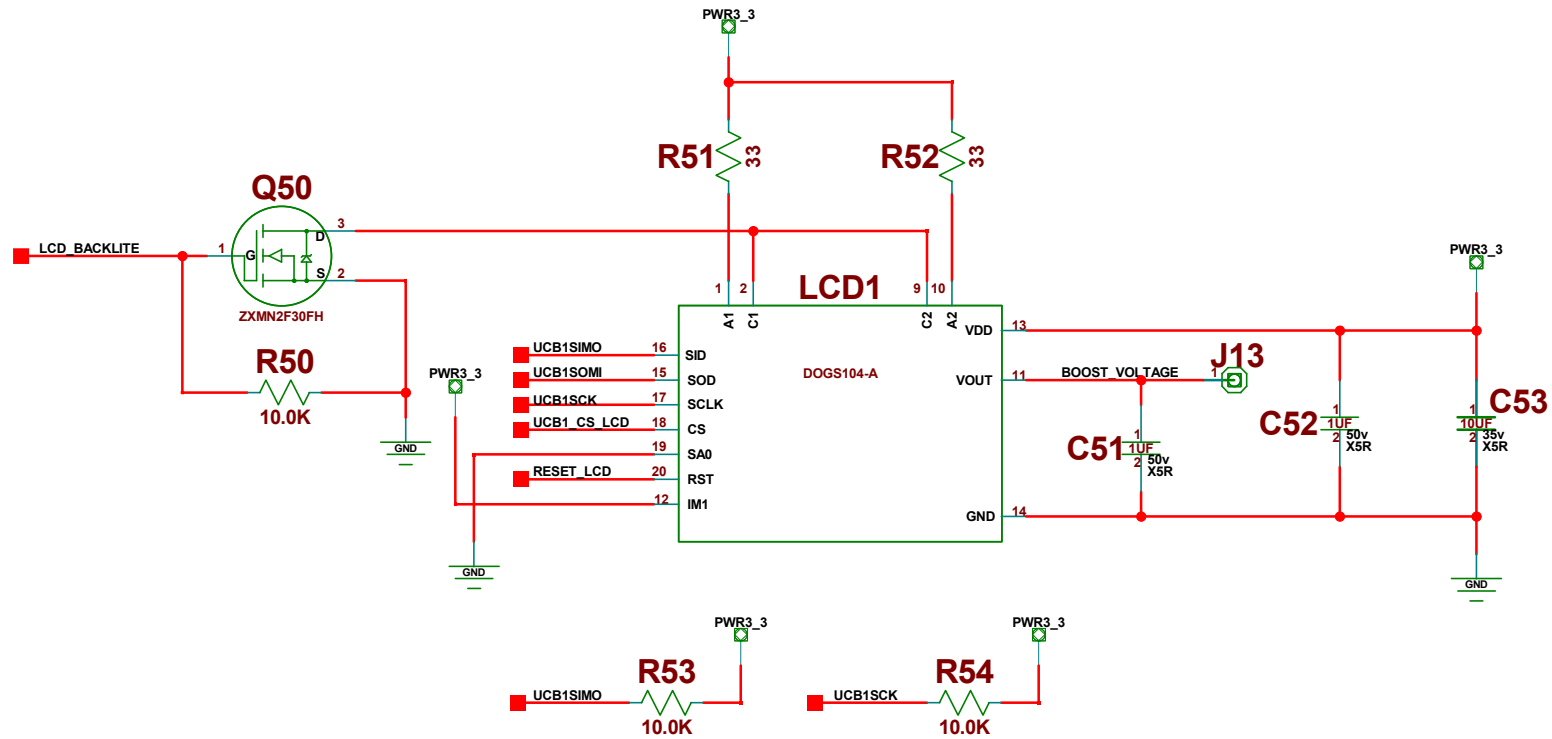


Center Emitter



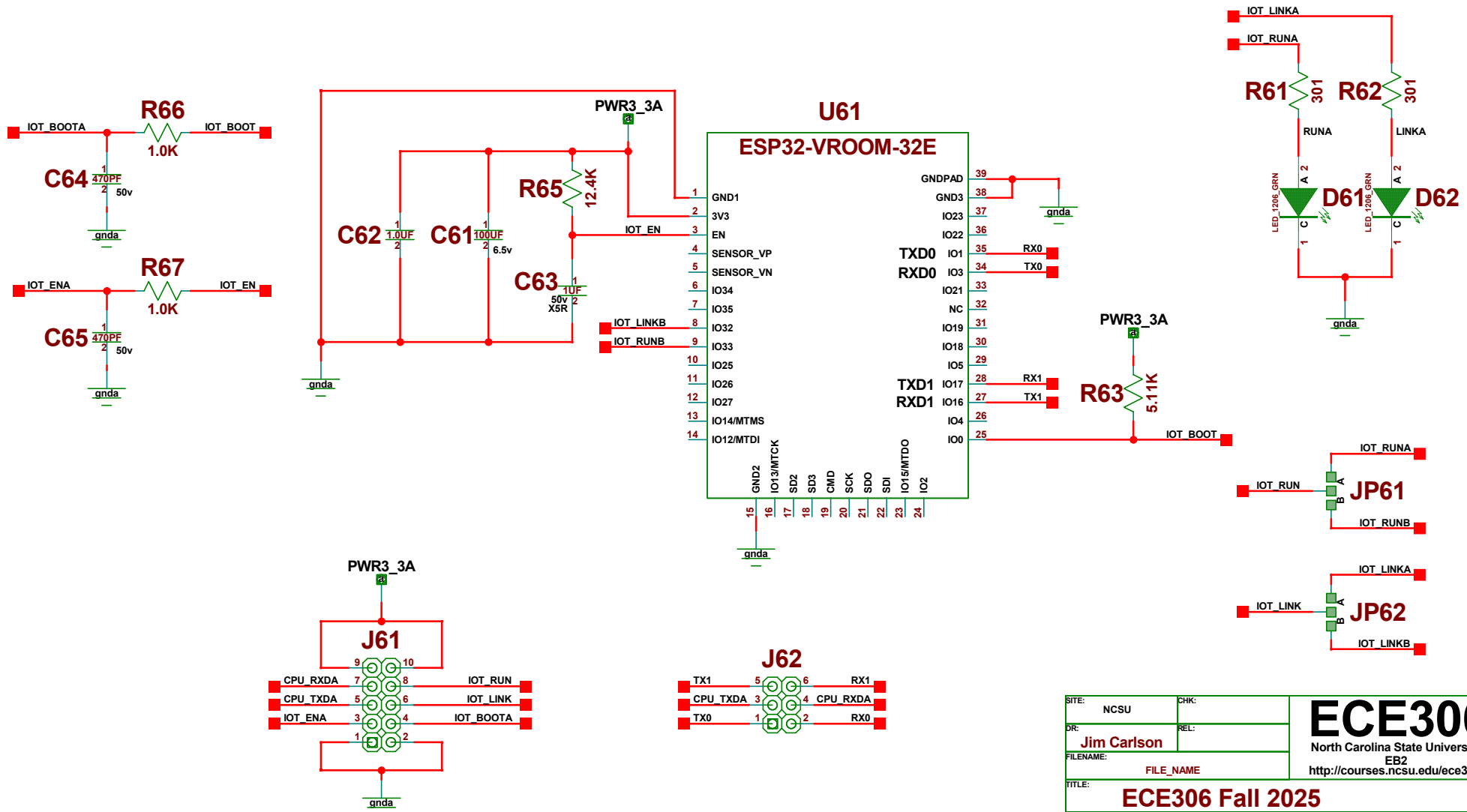
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1:1	B	ECE306		1 OF 1	5 OF 8	01
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LCD



SITE: NCSU		CHK:		ECE306 North Carolina State University EB2 http://courses.ncsu.edu/ece306/	
DR: Jim Carlson		REL:			
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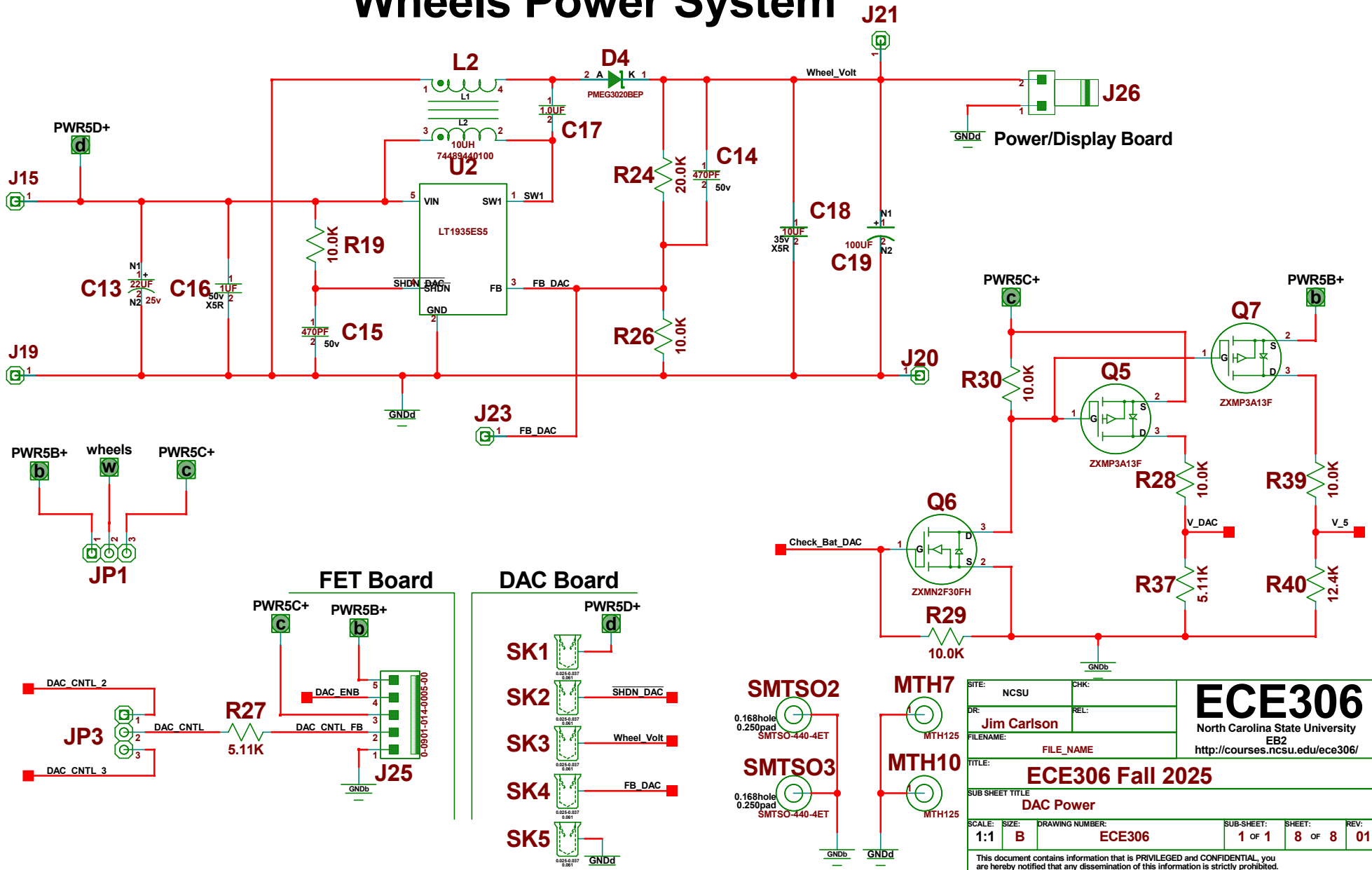
ESP32-VROOM



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DR: Jim Carlson		REL:		
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TITLE: ECE306 Fall 2025				
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Wheels Power System



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DR: Jim Carlson		REL:	
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